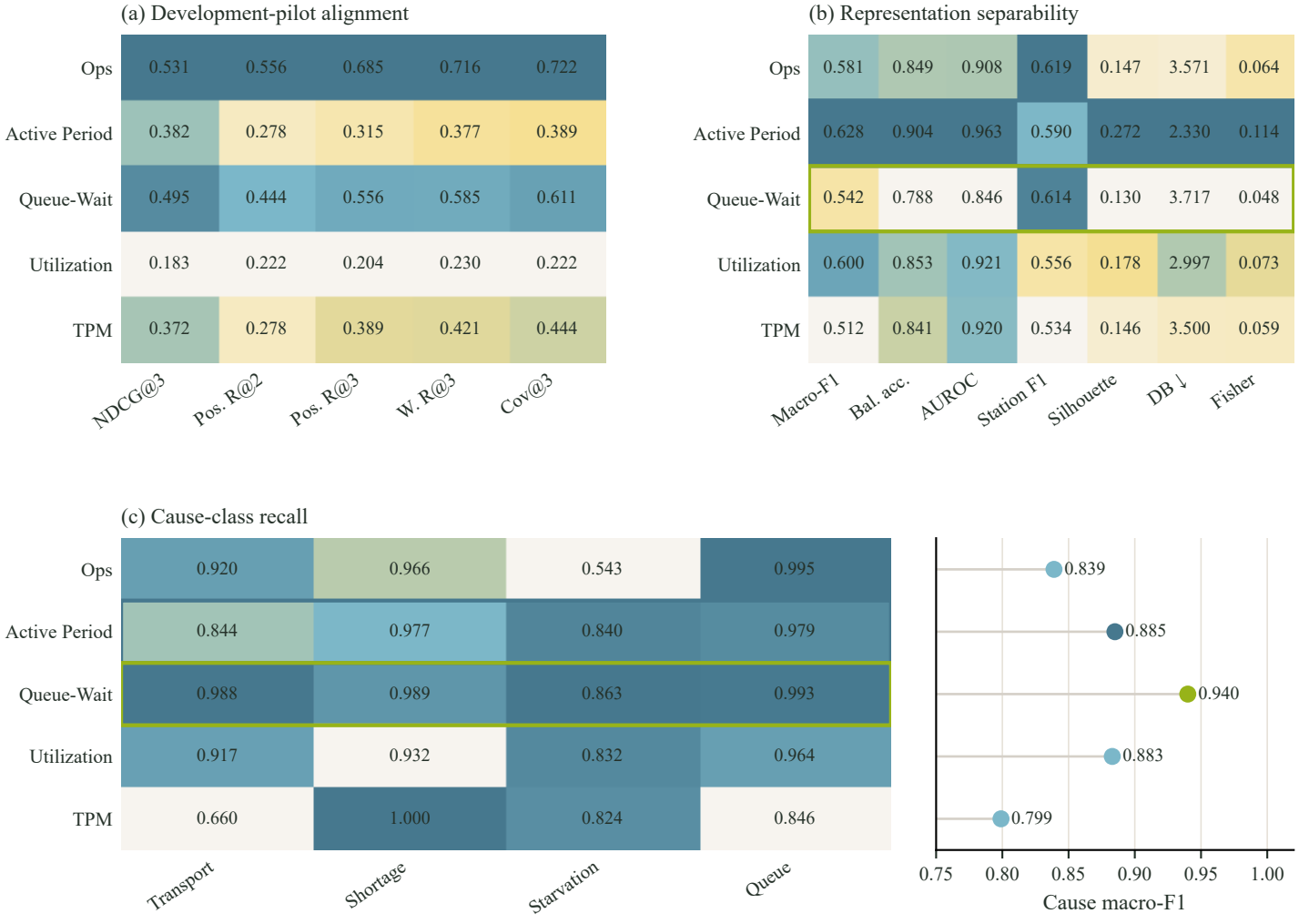
(a) Development-pilot alignment
Ops
Active Period
Queue-Wait
Utilization
TPM
0.531 0.556 0.685 0.716 0.722
0.382 0.278 0.315 0.377 0.389
0.495 0.444 0.556 0.585 0.611
0.183 0.222 0.204 0.230 0.222
0.372 0.278 0.389 0.421 0.444
NDCG@3
Pos. R@2
Pos. R@3
W. R@3
Cov@3
(b) Representation separability
Ops
Active Period
Queue-Wait
Utilization
TPM
0.581 0.849 0.908 0.619 0.147 3.571 0.064
0.628 0.904 0.963 0.590 0.272 2.330 0.114
0.542 0.788 0.846 0.614 0.130 3.717 0.048
0.600 0.853 0.921 0.556 0.178 2.997 0.073
0.512 0.841 0.920 0.534 0.146 3.500 0.059
Macro-F1
Bal. acc.
AUROC
Station F1
Silhouette
DB ↓
Fisher
(c) Cause-class recall
Ops
Active Period
Queue-Wait
Utilization
TPM
0.920 0.966 0.543 0.995
0.844 0.977 0.840 0.979
0.988 0.989 0.863 0.993
0.917 0.932 0.832 0.964
0.660 1.000 0.824 0.846
Transport
Shortage
Starvation
Queue
0.839
0.885
0.940
0.883
0.799
0.75 0.80 0.85 0.90 0.95 1.00
Cause macro-F1